

Erika Alexandra Hernández Zurilla

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EDUCATION

Universidad Metropolitana <i>Systems Engineering</i> • Minor in Software Development • Minor in Data Analysis and Processing	2022 – present Caracas, Venezuela
Universidad Central de Venezuela – Luis Razetti School of Medicine <i>Doctor of Medicine (M.D.)</i>	2013 – 2020 Caracas, Venezuela

PROFESSIONAL EXPERIENCE

Data Engineer Intern / Software Engineer Intern - Intelcon System <i>Consultant for VNET</i> • Designed and deployed automated n8n workflows processing a 200,000 subscriber base, ingesting ~1,000 new records monthly to enrich telecommunications network data end-to-end. • Integrated Python AI models to normalize, validate, and enrich infrastructure records, cutting manual processing. • Built backend services with NestJS, PostgreSQL, and TypeORM to power production automation pipelines. <i>Hydrocarbon Leak Detection Project</i> • Developing Python numerical models (RTTM with FiPy) to monitor a 10 km pipeline in real time, solving transient-flow PDEs via finite volume simulation.	05/2026 – Present Caracas, Venezuela
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ACADEMIC EXPERIENCE

Database I & Database II Teaching Assistant <i>Universidad Metropolitana</i> • Assisted professors in the creation of instructional materials and grading of exams. • Provided technical support in Neo4j, MongoDB, NestJS, Pentaho PDI, and PostgreSQL.	2022 – present Caracas, Venezuela
Software Engineering Teaching Assistant <i>Universidad Metropolitana</i> • Assisted in planning course activities and participated in Scrum ceremonies. • Acted as Scrum Master for 4+ teams per trimester. • Provided technical support in React, NestJS, and NextJS.	2022 – present Caracas, Venezuela

PROJECTS

YouTube Channel – “ErikaAlexandraH” <i>Academic and Technical Content</i> https://www.youtube.com/@erikaalexandrah  • Personal open-access project focused on teaching databases (MongoDB, Neo4j, PostgreSQL), backend development frameworks, ETL tools, and data analysis.	2023 – present
Web System for Separation Processes Laboratory <i>Universidad Metropolitana</i> • Developed a platform for managing laboratory reservations and inventory currently used by 100+ students and 5+ professors. • Built with NextJS and a backend integrated with Google Sheets through an API.	2023 – present
3D Interactive Virtual Campus in a Metaverse Environment <i>Universidad Metropolitana</i> • Award-winning project in the course <i>Digital Content for the Metaverse</i> (Unity). • Pioneer project for the development of the Teaching Assistant Program at Centro Mundo X.	2023

HONORS & AWARDS

Academic Excellence Recognition <i>Universidad Metropolitana</i>	2023
Cum Laude Distinction <i>Universidad Central de Venezuela</i> • Awarded upon graduation as Doctor of Medicine with outstanding academic performance. • Ranked 6th out of 169 graduates in the Luis Razetti School of Medicine class.	2020

SKILLS

Front-End Development – HTML, CSS, JavaScript, React, Next.js, Astro, Tailwind, Figma.
Back-End & Databases – Node.js, NestJS, PostgreSQL, MongoDB, Neo4j, Firebase, DynamoDB, Amazon Redshift, PDI (ETL).

LANGUAGES

Spanish Native	English Intermediate (B1)	German Basic (A2)
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